

# KAUSTUBH TRIPATHI

AI Engineer

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Backend engineer (1+ yr, Java & Spring Boot at Tata Consultancy Services) moving into AI engineering, backed by Anthropic's Claude Certified Architect and Claude Certified Developer credentials. Built two production-style LLM systems: a 6-route, multi-model GPT-5 pipeline with structured outputs and safety guardrails for a hackathon fraud-support platform, and an autonomous Gemini function-calling agent with full decision auditability. Strong CS fundamentals, REST/microservices background, and end-to-end full-stack delivery experience.

## CERTIFICATIONS

- Claude Certified Architect – Foundations — Anthropic, 2026 — [Verify](#)
- Claude Certified Developer – Foundations — Anthropic, 2026 — [Verify](#)

## TECHNICAL SKILLS

**AI / LLM Engineering:** LLM API Integration (OpenAI GPT-5 family, Google Gemini), Agentic Workflows & Function/Tool Calling, Prompt Engineering, Structured Output & JSON-Schema Enforcement, Multi-Model Routing, AI Safety Guardrails & Fallback Design, LLM Observability & Auditability

**Languages & CS Fundamentals:** Java, TypeScript, JavaScript, SQL, C++, Data Structures & Algorithms, Object-Oriented Design

**Frameworks:** Spring Boot, Spring Security (JWT, OAuth2), Node.js, Express.js, React.js, Next.js, Tailwind CSS, WebSockets (Socket.io), REST APIs, Microservices

**Databases:** PostgreSQL, MongoDB (Geospatial Indexing), Redis, MySQL, Supabase

**Tools & DevOps:** Docker, Docker Compose, Git, GitHub Actions, Jenkins, Postman, Vitest, Linux

## EXPERIENCE

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| <b>Tata Consultancy Services Ltd.</b><br>Product Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Bangalore, Karnataka<br>June 2025 – Present |
| <ul style="list-style-type: none"><li>Built and enhanced backend modules for a fintech/banking product in Java and Spring Boot, integrating new functionality into legacy systems while maintaining scalability and stability.</li><li>Optimized database performance by identifying bottlenecks and applying SQL query tuning and indexing, improving execution efficiency of critical processes.</li><li>Collaborated cross-functionally to ship features and fixes, participating in code reviews and maintaining coding standards; supported build automation via Jenkins to minimize regressions.</li><li>Owned debugging of production issues — analyzing logs and tracing API failures — to reduce issue resolution time.</li></ul> |                                             |

## PROJECTS

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| <b>Kavach — AI Cybercrime Support Platform (Build What Moves India Hackathon)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">Live Demo</a>   <a href="#">GitHub</a> |
| <ul style="list-style-type: none"><li>Co-built (2-person team) an AI-assisted cybercrime-reporting platform that turns a fragmented, multi-agency filing process into a single case file with ten ordered, deadline-tracked action steps.</li><li>Designed a 6-route LLM pipeline (triage, identifier extraction, document drafting, translation, transcription, grounded Q&amp;A) across GPT-5, GPT-5-mini and GPT-4o-transcribe, using strict JSON-schema-enforced structured outputs.</li><li>Built a provider-agnostic HTTP client against the OpenAI-compatible API (no SDK lock-in), a hybrid regex + LLM extraction pipeline, and a contradiction-detection guardrail that rejects AI-drafted output conflicting with a citizen's confirmed facts.</li><li>Shipped full graceful degradation — every AI route falls back to a deterministic rules engine so the app works with zero API key — plus support for all 22 scheduled Indian languages and local evidence storage with SHA-256 fingerprinting.</li></ul> |                                                    |
| <b>LeetCode Agent Tracker — Autonomous Interview-Prep Platform</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <a href="#">GitHub</a>                             |
| <ul style="list-style-type: none"><li>Built a full-stack interview-prep platform (React, Express, MongoDB) with a Leitner-style spaced-repetition scheduler and an analytics dashboard covering topic heatmaps and company readiness.</li><li>Designed an autonomous weekly-planning agent (Google Gemini function calling) that independently selects from 4 database tools across up to 8 iterations and returns a rationale-backed study plan.</li><li>Built a Socratic hint tutor that withholds solutions via graduated multi-turn hints, plus a post-mortem agent that reviews solved code to flag suboptimal time/space complexity.</li><li>Logged every agent invocation — tool calls, reasoning trace, latency — to a queryable collection for full decision auditability.</li></ul>                                                                                                                                                                                                                             |                                                    |
| <b>Skill Barter Platform — Peer-to-Peer Skill Exchange</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">Demo</a>   <a href="#">GitHub</a>      |
| <ul style="list-style-type: none"><li>Engineered a matching engine classifying users into mutual vs. one-way matches, ranked by geospatial proximity via MongoDB 2dsphere indexes, with a real-time Socket.io chat and a two-way, mutual-consent trade lifecycle.</li><li>Implemented dual-token JWT authentication with silent renewal and OTP email verification; deployed on Render/Vercel with production-grade security (bcrypt password hashing, CORS whitelisting, rate limiting).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                    |

## EDUCATION

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| <b>Bachelor of Technology, Computer Science and Engineering</b><br>Madan Mohan Malaviya University of Technology, Gorakhpur, Uttar Pradesh — CGPA: 7.9 | Oct 2020 – June 2024 |
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